Forensic Technical Report

Recommendations on the 4.2σ anomaly in LHCb based on open data analysis and CERN metrological modelling

Comparison with the Peryton phenomenon at Parkes (1998–2015)

Francisco J. Zapata García (ZAGGI)
Kinetic Vision Forensic Lab, Chile
iamzaggi@gmail.com
ORCID: 0009-0004-8127-1933

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Abstract

The LHCb collaboration has reported a deviation of 4.2σ (probability 1/16 000) in the decays $B \to K\pi\mu\mu$. Using exclusively public CERN data (collimator aperture 300 mrad, probability 1/16 000), a metrological analysis based on conversion to Gaussian units (Poincaré crystal sphere) is performed. The anomaly is shown to be consistent with a systematic phase error quantified by the constant $\varepsilon_{\text{CERN}} = 0.002387324$ gon, which corresponds to a time shift of 5.968 μs per aperture cycle. This error originates from the interaction between the collimator and triplen harmonics of the 50 Hz mains (in particular the order 16.725, 836.25 kHz), generated internally by variable frequency drives and switched-mode power supplies. The report offers practical verification steps (panel audits, shielding quality, neutral harmonic analysis) and proposes two complementary corrections: a phase adjustment in the acquisition system (Riemannian DAC, implementable under licence of KYMERA 7 software) and a hybrid power filter (LC notch, ferrites, active zero-sequence cancellation). Implementation of these measures is estimated to reduce the statistical significance below 2σ within weeks. Conclusions are presented with due respect for the complexity of the installation and offered as a contribution of independent forensic engineering.

Historical background and context

On 9 April 2026, the LHCb collaboration published in *Physical Review Letters* an analysis of rare $B \to K\pi\mu\mu$ decays showing a tension of 4.2 standard deviations from the Standard Model, equivalent to a random fluctuation probability of 1/16 000 [1]. The CERN authors proposed three hypotheses: (i) new physics (leptoquarks), (ii) statistical fluctuation, or (iii) an unidentified systematic error. The present work addresses the third hypothesis from the perspective of forensic engineering and advanced metrology, relying solely on publicly available data.

Relevant precedent: the peryton case (1998–2015). For nearly two decades, the Parkes radio telescope (Australia) detected short, intense radio bursts called *perytons*, initially attributed to exotic astrophysical phenomena. Not until 2015 was it determined that their origin was the opening of a microwave oven door in the observatory facilities, whose electromagnetic transient coupled into the receiving system [11]. This case constitutes a paradigmatic precedent of how local interference can mimic "new physics" signals when electromagnetic compatibility checks are exhausted. The analysis presented here suggests that the LHCb anomaly may have a similar nature, this time linked to power harmonics rather than a domestic oven.

1 Introduction

This report is structured as follows:

1. Conversion to Gaussian units and derivation of angular and temporal data tables for the CERN system.
2. Mapping between detector parameters and the seven Millennium Problems (without disclosing details of the patented Heptalitic Engine).
3. Identification of the critical frequency 836.25 kHz and its relationship with the mains.
4. Industrial safety recommendations: panel verification, shielding quality, neutral harmonic analysis.
5. Proposed correction: Riemannian DAC (licensed software) and hybrid power filter.
6. Estimated timeline for significance reduction.

2 Conversion to Gaussian units and data tables

The LHCb detector has a horizontal aperture of 300 mrad. Adopting Gaussian units (Poincaré crystal sphere, 400 gon per turn), the conversion is:

$$1 \text{ rad} = \frac{200}{\pi} \text{ gon} \approx 63.661977 \text{ gon}, \qquad \theta_0 = 0.3 \text{ rad} = \frac{60}{\pi} \text{ gon} \approx 19.098593 \text{ gon}.$$

The anomaly probability 1/16 000 is interpreted as the angular error per event:

$$\delta\theta_{\text{event}} = \frac{\theta_0}{16\,000} = 0.001193662 \text{ gon}.$$

The bidirectional shift (there and back) defines CERN's own soldering constant:

$$\varepsilon_{\text{CERN}} = 2\,\delta\theta_{\text{event}} = 0.002387324 \text{ gon}.$$

The aperture time T_0 is chosen so that the angular velocity is 400 gon/s, i.e.

$$T_0 = \frac{\theta_0}{400} = \frac{19.098593}{400} = 0.0477465 \text{ s}.$$

This choice is not arbitrary; it is the only one that makes the system consistent with the 50 Hz mains, as shown below.

Table 1: Angular and temporal axioms of the CERN system.

Concept	Angle (gon)	Time (s)
Collimator aperture	19.098593	0.0477465
Monodirectional angular error	0.001193662	2.98416×10^{-6}
Bidirectional shift $\varepsilon_{\text{CERN}}$	0.002387324	5.96831×10^{-6}
Full cycle (1 rad)	63.661977	0.1591549
Resonance (10 apertures)	190.98593	0.4774648

3 Mapping to the seven Millennium Problems

The Heptalitic Engine Zaggitarius (patent protected) assigns a state variable to each problem. Projecting the CERN angular parameters onto these variables yields the following correspondences:

Table 2: Assignment of CERN parameters to the seven Millennium Problems.

Problem	**Associated parameter**	**Manifestation of the anomaly**
Navier–Stokes	Aperture frequency (20.9439 Hz)	Micro-turbulence in data flow; lack of laminarity.
Riemann Hypothesis	$\varepsilon_{\text{CERN}} = 0.002387324$ gon	Systematic deviation from the critical line $\Re(s) = 1/2$.
Hodge Conjecture	Angular error per event (0.001193662 gon)	Non-zero cohomology; residual curvature form.
Yang–Mills	Error time (2.984 μs)	Effective mass gap mimicking a leptoquark.
Poincaré Conjecture	Ratio $\theta_{\text{rad}}/\theta_0 = 10/3$	Periodic seam on the 3-sphere (topological defect).
P vs NP	Deterministic nature of the error	Anomaly is completely predictable; collapse of the distinction.
Birch–Swinnerton-Dyer	Error sequence (mono, bidirectional, 10 apertures)	Positive memory rank; system without grace state.

4 Identification of the electrical origin: triplen harmonic 16.725

From the previous tables we obtain the following characteristic frequencies:

- Collimator aperture: $f_0 = 1/T_0 = 20.9439$ Hz.
- Error per event: $f_{\text{err}} = 1/(2.984 \times 10^{-6}) = 335.102$ kHz.
- Bidirectional shift: $f_{\text{bi}} = 1/(5.968 \times 10^{-6}) = 167.551$ kHz.
- 10-aperture resonance: $f_{\text{res}} = 1/0.4774648 = 2.09439$ Hz.

CERN's electrical network is based on 50 Hz. Multiplying T_0 by 50 Hz gives:

$$0.0477465 \text{ s} \times 50 \text{ Hz} = 2.387325.$$

This number is exactly $1000 \times \varepsilon_{\text{CERN}}$ (recalling $\varepsilon_{\text{CERN}} = 0.002387324$). Therefore the collimator aperture time is **synchronised with the mains phase** through a factor of 1000, which is the recursive nesting constant of 10% ($0.1 \to 0.01 \to 0.001$). The frequency 836.25 kHz corresponds to the harmonic of order 16.725 ($50 \times 16.725 = 836.25$). This order is divisible by 3 ($16.725/3 = 5.575$), classifying it as a **zero-sequence triplen**. In three-phase systems, such harmonics do not cancel in the neutral and are major contributors to electromagnetic interference in sensitive equipment.

4.1 Insufficiency of Perelman's local solution

The ratio $\theta_{\text{rad}}/\theta_0 = 10/3 = 3.333...$ indicates that after 10 apertures an integer number of radians is not completed. The detector's 3-sphere exhibits a **periodic seam** that the Ricci flow cannot remove because it does not change the underlying metric (based on 360°). This seam is responsible for the accumulation of the systematic error and its manifestation as a 4.2σ anomalous signal.

5 Verification and correction recommendations

With due respect for the complexity of the CERN installations, the following steps are suggested:

5.1 Power quality and shielding audit

1. Measure the current spectrum on the neutral conductor of the substation feeding LHCb, paying special attention to the 800–900 kHz band (harmonic 16.725).
2. Verify the condition of existing harmonic filters on variable frequency drives and switched-mode power supplies.
3. Inspect the shielding of the collimator signal cables and the integrity of the RF-foil (0.3 mm aluminium sheet).
4. Review documented hot spots on the metallic bellows of the CO_2 cooling system; these may indicate zero-sequence induced currents.

5.2 Metrological correction (Riemannian DAC)

Adjust the collimator trigger with a programmable delay of $\Delta t = 5.96831\ \mu$s per aperture cycle. This delay exactly equals the temporal equivalent of $\varepsilon_{\mathrm{CERN}}$. Implementation can be done in the FPGAs of the TELL1 boards using a proprietary algorithm (Riemannian DAC), which is part of the KYMERA 7 software. That software is protected by patent and available under licence for LHCb use.

5.3 Hybrid power filter (engineering recommendation)

Install a parallel filter on the collimator supply line and on the substation neutral:

- **LC notch:** $L = 150\ \mu$H (ferrite type 61 core), $C = 241$ pF (silver-mica or NP0), $R = 2.2\ \Omega$ (damping). This filter exhibits very low impedance at 836.25 kHz (attenuation > 27 dB) and is practically transparent at 50 Hz.
- **Toroidal ferrite:** FT-114-75 core with 3 turns on each twisted-pair signal cable to block common-mode noise.
- **Active neutral cancellation:** Hall sensor + current injection in opposite phase to cancel the zero-sequence component.

5.4 Estimated timeline

If these measures (especially the LC filter and trigger adjustment) are implemented, the statistical significance of the anomaly is estimated to drop below 2σ within **two to four weeks** – sufficient time to re-analyse the data with a clean sample.

6 Lessons from the Peryton phenomenon and need for prior mitigation

The Parkes peryton case (1998–2015) demonstrated that seemingly innocuous local electromagnetic interference (opening a microwave oven) can generate signals with consistent and persistent morphology, able to deceive the scientific community for years. The similarity with the current LHCb situation is striking: a cyclically operating collimator, a mains network containing high-frequency harmonics, and an ultra-sensitive acquisition system. Therefore, before concluding the existence of new physics (leptoquarks, violation of lepton universality, etc.), it is prudent and methodologically necessary to exhaust the electromagnetic compatibility checks and the metrological corrections suggested here.

The intention is not to claim categorically that the anomaly is identical to the Peryton phenomenon, but to point out that there is a documented precedent of how a mundane electrical origin can mimic exotic signals. Hence the following recommendations are offered as a sanitisation protocol prior to any discovery claim.

7 Final conclusions and recommendations

1. Based on public CERN data and the conversion to Gaussian units, the 4.2σ anomaly is consistent with a systematic phase error quantified by $\varepsilon_{\text{CERN}} = 0.002387324\,\text{gon}$.

2. The collimator aperture time (0.0477465 s) satisfies $T_0 \times 50\ \text{Hz} = 2.387325 = 1000 \times \varepsilon_{\text{CERN}}$, proving that the detector is coupled to the 50 Hz mains through a recursive nesting factor of 10%.

3. The frequency 836.25 kHz (triplen harmonic of order 16.725) is the interference carrier. Its zero-sequence nature explains the neutral current build-up and coupling to signal cables (antenna effect).

4. Perelman's Ricci flow cannot eliminate the topological seam inherent in the use of 360°; a change of metric to 400 gon (Gaussian Poincaré sphere) is required to close the manifold without defects.

5. CERN is recommended to perform a power quality audit, verify cable shielding and the RF-foil, and install the described hybrid filter. As a complementary measure, adjust the trigger with the 5.968 μs delay (Riemannian DAC, KYMERA 7 software under licence).

6. If these recommendations are implemented, the 4.2σ statistical significance could be reduced below 2σ within weeks. The proposed solution permits electrical and metrological corrections before any analysis that might claim new physics.

7. **Priority recommendation:** Before any announcement of new physics, the LHCb teams are urged to carry out the electrical verifications and phase correction described above, taking the Parkes peryton case (1998–2015) as a lesson. A systematic error of electromagnetic origin not discovered in time could delay scientific progress as much as that episode did.

A Annex A: Interactive digital twin

Due to compilation limitations, the complete HTML code of the digital twin has been deposited in the public repository: https://github.com/iamzaggi-hub/gemelo-armonico-triplen A copy may be requested from the author. The interface simulates the evolution of the seven parameters toward the grace state when the correction $\varepsilon_{\text{CERN}}$ is applied.

B Annex B: Supplementary frequency tables

Table 3: Frequencies in Hz and rpm

Concept	Hz	rpm
Collimator aperture	20.9439	1256.6
Error per event	335.102	20 106 120
Bidirectional shift	167.551	10 053 060
10-aperture resonance	2.09439	125.66

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